

Final Project – Individual Contribution Statement

Agri-DINO: Domain-Adaptive Self-Supervision for Fine-Grained Plant Recognition

Md Mehedi Hasan
Auburn University

I contributed equally to the project. My primary responsibility was downstream evaluation, benchmark validation, and the second-pass manuscript draft.

Code and experiments

- Prepared the MedicinalPlant and PlantSeg layouts, verified the stratified train/validation split, and organized the deterministic evaluation harness.
- Implemented and ran the random-initialization and ImageNet-21k baseline configurations, documenting the difference between direct fine-tuning and Agri-DINO’s LP-FT protocol.
- Implemented the cumulative ablations for resolution, LP warmup, LLRD, and horizontal-flip TTA.
- Audited the PlantSeg YOLO-to-classification conversion, traced the sample-level filename parsing, and identified the inflation from 20 disease categories to 300 micro-classes.
- Debugged evaluation scripts and checked class counts, labels, preprocessing, input resolution, and metric computation.

Analysis and tables

I ran the baseline, ablation, and PlantSeg audit experiments. I analyzed validation accuracy, validation loss, training–validation gaps, and the LP-to-LLRD transition; prepared the main-results, ablation, training-dynamics, and PlantSeg tables; and checked the arithmetic and consistency of the reported values, captions, and claims.

Writing and shared work

I drafted Related Work, Experiments, the PlantSeg analysis, and portions of the Discussion and Conclusion. I also edited the full manuscript for readability, protocol clarity, and reproducibility. With Md Min-Ha-Zul Abedin, I reviewed the literature, verified the final numbers, debugged the pipeline, and checked the final LaTeX/PDF output.